

JUNIOR CERTIFICATE

CHEMISTRY

Form 2

Study Notes

Table of Contents

TOPIC 1: ELEMENTS AND THE PERIODIC TABLE	5
Blocks of elements in the periodic table	5
The general trends of the periodic table	5
Physical and chemical properties of the families of elements	7
a. Group I elements - Alkali metals	7
Physical properties of group one elements.....	7
Chemical properties of alkali metals.....	8
Uses of alkali metals	9
b. Group II Elements-Alkaline earth metals	10
Physical properties of alkaline earth metal as you go down the group	10
Chemical properties of alkaline earth metals	10
The uses of Alkaline-earth metals	10
c. Group VII elements - Halogens	11
Physical properties of group 7 elements	11
Chemical properties of group 7 elements.....	11
Uses of Group VII elements.....	12
d. Group VIII elements – Noble gases	12
Physical properties of noble gases	13
Uses of group 8 elements.....	13
TOPIC 2: CHEMICAL BONDING	14
Types of chemical bonding	14
Ionic bonding	14
Properties of ionic bonding	16
Covalent Bonding	17
Properties of covalent bonding	18
Metallic bonding	18
Physical properties of metals	18
TOPIC 3: ACIDS AND BASES	20
Properties of acids and bases	20
Electrical conductivity of acids.....	20
Reactions of acids	21
Uses of acids	21
Properties of bases	22
Uses of bases.....	22

Acid-Base Indicators.....	22
Types of indicators.....	23
Neutralisation.....	25
Application of neutralisation reaction.....	25
TOPIC 4: HYDROCARBONS.....	26
Families of organic compound.....	26
Homologous series.....	26
Characteristics of homologous series.....	26
Saturated and Unsaturated Hydrocarbons.....	26
Alkanes	26
Naming of alkanes	27
The four types of formulae of alkanes	27
Sources of alkanes.....	28
Fractional distillation	29
Cracking of alkanes.....	29
Physical properties of alkanes.....	29
Chemical properties of alkenes	30
Uses of alkanes	30
Alkenes	31
Naming of alkenes	31
Sources of alkenes.....	32
Physical properties of alkenes.....	33
Chemical properties	33
Uses of alkenes	33
TOPIC 5: AIR	34
What is air?	34
Separating component of air using fractional distillation of liquid air	34
Uses of gases found in air	35
Quantitative determination of the fraction of oxygen in air.....	35
Physical properties of oxygen.....	36
Chemical properties of oxygen	37
Uses of oxygen.....	37
Atmospheric pollutants and their sources	37
Poisonous gases and their effects on health and environment	37
Sources of pollutants.....	38
Effort made to reduce atmospheric pollution.....	38
TOPIC 6: SOIL	39

What is soil?.....	39
Soil composition	39
Chemical composition of soil.	39
Soil pollution.....	40
Prevention of soil pollution.....	40

TOPIC 1: ELEMENTS AND THE PERIODIC TABLE

Groups	1	II	III	IV	V	VI	VII	VIII
	1H 1 1							4He 2 2
	7Li 3 2,1	9Be 4 2,2	11B 5 2,3	12C 6 2,3	14N 7 2,4	16O 8 2,6	19F 9 2,7	20Ne 10 2,8
	23Na 11 2,8,1	24Mg 12 2,8,2	27Al 13 2,8,3	28Si 14 2,8,4	31P 15 2,8,5	32S 16 2,8,6	35 Cl 17 2,8,7	40Ar 18 2,8,8
	39K 19 2,8,8,1	40Ca 20 2,8,8,2						

Blocks of elements in the periodic table

Elements in the periodic table are divided into three. Namely; metals. Non-metals and metalloids.

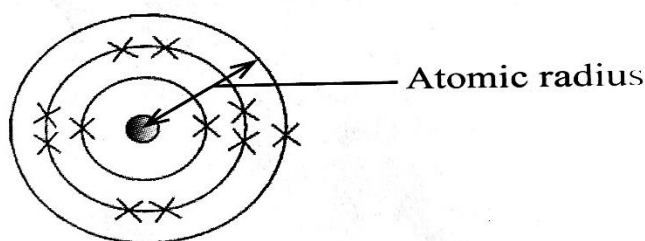
- **Metals** are mostly found in group one, two and three of the periodic table. E.g. Lithium, Magnesium, calcium etc.
- **Metalloids**, these are elements that elements that show properties partly to those of metals and partly to those of non-metals.
- **Non-metals and transition metals**, are commonly found in group four, five, six, seven and eight. E.g., nitrogen, oxygen, chlorine argon etc.

The general trends of the periodic table

1. Atomic radius

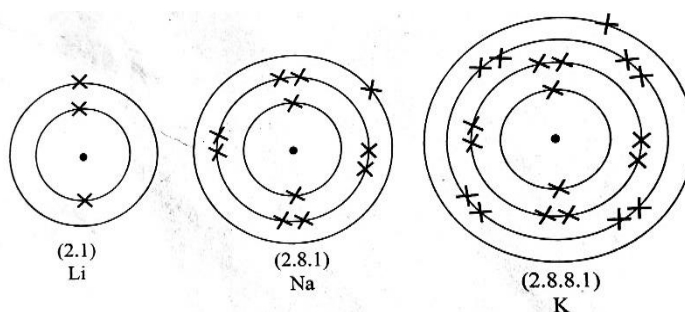
- The atomic radius is the distance between the nucleus and the outermost energy level in an atom.

- The more the number of energy levels, the greater the atomic radius and vice versa. .g



Atomic radius of sodium

- Consider lithium (Li), sodium (Na) and potassium (K) with 3, 11 and 19 electrons respectively.
- Their electronic configurations and atomic structures will be as follows;



- From the figures about you note that the distance from the nucleus to the outermost energy level increases as we move from lithium to potassium. That is, the atomic radius is increases.
- From the three Group I elements, lithium has the smallest atomic radius followed by sodium. Potassium has the largest atomic radius.
- The atomic radii increase as we go down the group as shown above.
- The shielding effect by the electrons on the positive nucleus affects the size of the atom.
- **Shielding effect or screening effect** refers to the reduced action by the nucleus for the outermost electrons caused by the electrons within the inner energy levels.

2. Ionisation energy

- Ionisation energy is that energy, that must be absorbed in order to remove the outermost electron from an atom.
- The higher the ionisation energy the more difficult it is to remove an electron from an atom.

Element	Ionization energy (K)	Atomic radius
Lithium (Li)	520	0.15 um
Sodium (Na)	496	0.19 um
Potassium (K)	479	0.23um

- Notice that the ionisation energy decreases as you move down the group while atomic radius increases. This means that it is easier to remove an electron from the atom as we go down the group 1
- The smaller the atom the higher the ionization energy. Electrons are more attracted to the nucleus in a smaller atom

3. Electron affinity

- **Electron affinity** is concerned with the attraction of electrons.
- An electron trying to enter into an atom will be repelled by the electrons in the outermost energy level.
- Since the atomic radii increase down a group of a Periodic Table, **electron affinity** decreases down a group,
- Electron affinity increases across a period across a period because the atomic radii decrease because electrons fill the same energy level and the nuclear charge (protons) increases.

4. Electronegativity

- Is the electron - attracting power of an atom
- Electronegativity increases across a period and decreases down a group of the Periodic.

Physical and chemical properties of the families of elements

a. Group I elements - Alkali metals

- Group one elements are the most reactive group of metals.
- They are found on the extreme left-hand side of the Periodic Table, members of this group include Lithium (Li), Sodium (Na) and Potassium (K).

Physical properties of group one elements

- They are soft metals and they can be easily cut by a knife
- They have low melting and boiling points as compared to other metals

- They are good conductors of heat and electricity
- They have low density and float on water
- They are shiny when freshly cut
- They have the largest atomic radii in their period

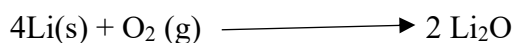
General trends of the physical properties of alkali metals

- The melting and boiling points decrease due to an increase in atomic size hence attraction between positive nucleus and the electrons as reduced.
- Atoms become bigger due to increase in number of energy levels.
- Density increases due to increased mass.
- Ionization energy decreases down due to increase in atomic size hence weaker forces of attraction between the possible nucleus and outermost energy level electrons.
- Electron affinity reduces as the atom becomes bigger.
- Electronegativity decreases due to increase in atomic size.

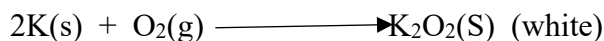
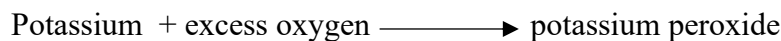
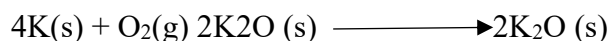
Chemical properties of alkali metals

1. Reaction with oxygen

- They burn in air to form oxide and peroxide depend on the amount of oxygen present
Example. Lithium + oxygen $\longrightarrow$ lithium Oxide

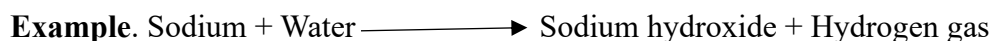


- Potassium also reacts with oxygen to form a mixture of oxide and peroxide



2. Reaction with water.

- Group one elements react with water to form salt solution and hydrogen gas

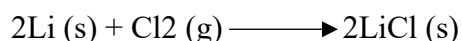


- The reaction is slowly for elements up the group and is very fast and explosively with those down the group e.g potassium is more reactive metal.

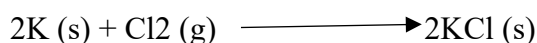
3. Reaction with chlorine

- Alkali metals react with chlorine to form chlorides

Example. Lithium + chlorine $\longrightarrow$ lithium chloride



Potassium + Chlorine $\longrightarrow$ potassium chloride



Uses of alkali metals

Lithium

- Used in production lubricating grease
- Used in deoxidizing copper and copper alloys.
- Lithium-6 is a main source for the production of titanium.
- Used in batteries which contain more energy compared to other metals. (batteries used in cell phones and computers)

Sodium

- Used in preparation of tetra-ethyl lead which is used as antiknock reagent in petrol. (Tetraethyl lead is being phased out to reduce lead pollution problem).
- Used in extraction of titanium metal from titanium chloride.
- Sodium vapour is used in lamps for streetlighting.
- Sodium chloride is used to add taste to food.
- Sodium carbonate is used in manufacturing of glass, detergents and for softening hard water.
- Sodium hydroxide and sodium chloride are used in manufacture of soap and detergents.

Potassium

- Potassium chloride is used in making fertilizers (Is essential for plant growth).

- Potassium chromate is used in tanning leather, in manufacture of inks, gun powder and dyes.
- Potassium nitrate is used as food preservative.

b. Group II Elements-Alkaline earth metals

- Found at the left-hand side of the periodic table just after Group I elements.
- Examples include; Beryllium (Be), Magnesium (Mg), Calcium (Ca)

Physical properties of alkaline earth metal as you go down the group

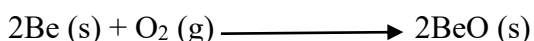
- Melting and boiling points decrease down the group. This is because of increased atomic size which leads to decrease in the forces of attraction between the positive nucleus and the electrons.
- Atoms become bigger due to increase in number of energy levels.
- Density increases due to increase in mass.
- Ionization energy decreases due to increase in atomic size which weakens force of attraction between the possible nucleus and outermost energy level electrons.
- Electron affinity reduces as the atom becomes bigger.
- Electro negativity decreases due to increase in atomic size.

Chemical properties of alkaline earth metals

1. React with air or oxygen.

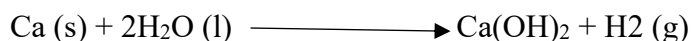
- They react with oxygen to produce oxides

Example; Beryllium + Oxygen $\longrightarrow$ Beryllium oxide



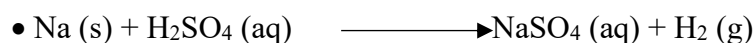
2. They react with water to form:

- **Example:** Calcium + Water $\longrightarrow$ Calcium hydroxide + Hydrogen gas



3. They react with dilute acids

- Example: Sodium + Sulphuric $\longrightarrow$ Acid Sodium Sulphate + Hydrogen gas



The uses of Alkaline-earth metals

Beryllium

- Used in transmission of X-rays.

- Alloy of beryllium and copper is hard strong with high resistance to wear hence used to make computer parts.
- Alloy of beryllium is used as structural materials for high performance air crafts, missiles, space craft and satellites.
- Beryllium oxide is used in nuclear industry.

Magnesium

- Used in making alloys used for manufacturing aircraft, parts of car engine casing and for missile construction.
- Used as a reducing agent in the production of uranium and other metals from salts.
- Used in pharmaceutical industry to manufacture medicine.
- Magnesium oxide is used in brick-liners in furnaces.
- Used in computer for radio-frequency shielding.

Calcium

- Used as reducing agent in the preparation of metals such as; Thorium, Uranium, zirconium
- Calcium form calcium carbonate which is a component of cement.
- Calcium carbonate is used in ant acid tablets.

c. Group VII elements - Halogens

- The word Hala -means salt and gen- means to produce, therefore halogens mean salt producing property.
- They are found at the right-hand side of the periodic table and exists as diatomic molecules.
- Elements in group seven include: Fluorine (F) chlorine (Cl) Iodine Bromine (I)

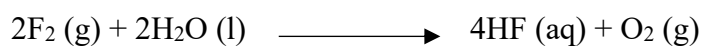
Physical properties of group 7 elements

- Fluorine and chlorine are gases at room temperature
- They have low melting and boiling point but increase as you go down the group
- All group seven elements act as diatomic molecules
- They do not conduct heat and electricity because they exist as molecules with strong intermolecular forces of attraction bonding the atoms together.

Chemical properties of group 7 elements

- They react with water to form hydrogen halide and oxygen.

Example; Fluorine + water $\longrightarrow$ Hydrogen fluoride + Oxygen



- They react with group one elements to form salt

Example; Chlorine + Sodium $\longrightarrow$ Sodium Chloride



- Halogens also react with other metals such as zinc, and Iron

Example Chlorine + Iron $\longrightarrow$ Iron III Chloride

Uses of Group VII elements

Fluorine

- Used in manufacture of plastics.
- Hydrofluoric acid is used in air conditioning and refrigerators.
- Small amount in water prevents tooth decay (for making toothpaste)

Chlorine

- Used in water treatment.
- For production of paper products dye stuffs textile petroleum medicine.

Bromine

- Used in making petrol engine anti-knock compounds.
- Used in the manufacture of Fumigants, Flame proof agents, Water purification compounds, Dyes, Medicine, pesticides

Iodine

- used in medicine such as disinfectant or external wounds is made from a solution of potassium iodide and iodine in alcohol.
- Silver iodide is used in photography.
- Is added to stable salt to prevent goiter
- Used in testing for starch in biological science.

d. Group VIII elements – Noble gases

- Elements in this group include; Helium (He), Neon (Ne), Argon (Ar)
- All elements in group eight are unreactive since their outermost shell is full.
- Helium is in group eight beside having two electrons in the outermost shell because its outer most shells is full and it is unreactive

Physical properties of noble gases

- They are all colorless gases
- They are all unreactive
- They do not conduct heat and electricity
- They exist as monoatomic gases at room temperature

Uses of group 8 elements

a) Helium

- It is used in filling meteorological balloons
- It is used as a cooling medium for nuclear reactors
- It is used in pressurizing rocket fuels to liquid
- Used as protective gas for semiconductors materials

b) Neon

- Used in making advertising colour sign
- Used to make high voltage indicators
- Used in making lasers
- Liquid neon is an economical refrigerant

c. Argon

- It is used in electric bulbs and fluorescent tubes
- Used as inert gas shield for arc welding and cuttings

TOPIC 2: CHEMICAL BONDING

- **Chemical bond** is a combination of atoms to attain stability
- **A chemical bonding** is the process whereby elements gain, lose or share electrons
- Is the union of two or more particles to form a new one or large particle than original.

Types of chemical bonding

- There are three main types of chemical bonding. Namely
 - Ionic bonding
 - Covalent bonding
 - Metallic bonding

Ionic bonding

- It is the electrical attraction between oppositely charged ions, which are produced when electrons are transferred from one atom to another during a chemical reaction.
- Elements attain stability by gaining or losing electrons so that their outermost shell has the highest number of electrons.
- Ionic bonding takes place between metals and non-metals e.g. Group 1, and 2 bonding with Group 6 and 7.
- Metals lose electrons and non-metals gain gas electron in order to be stable, that is, to reach an inert
- The atoms that lose electrons are positively charged while the atom that gains an electron are negatively charged

Valency

- Valency is the combining power of an ion or atom.
- The size of the charge of an ion is the measure of its valency or the combining power.

Valence electrons

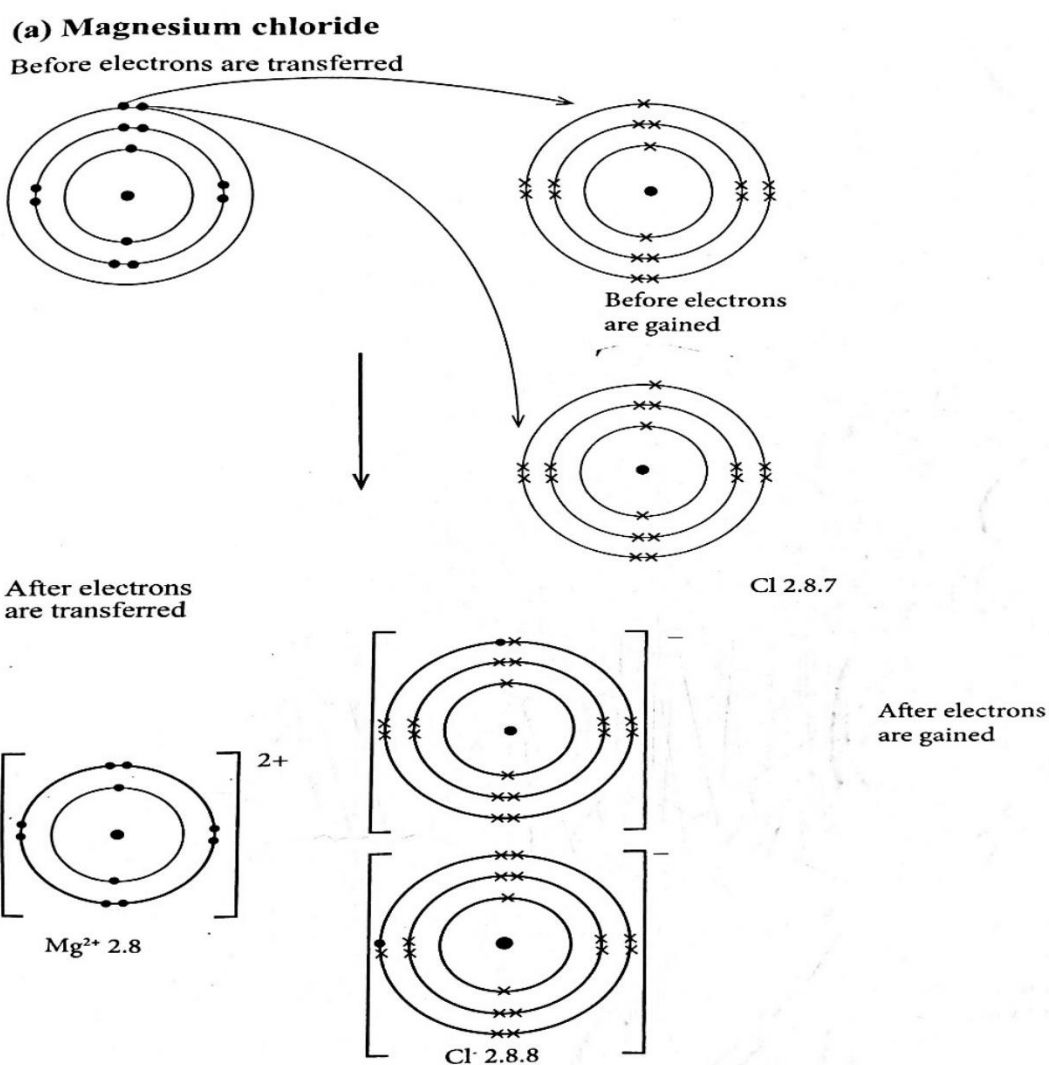
- It is the number of electrons found in the outermost shell.
- **Cation** is a positively charged ion for example Ca^{2+}
- **Anion** is a negatively charged ion for example Cl^-

Deducing molecular formulae

- When deducing a molecular formula make sure that the charges on the cations and anions cancel each other so that the total or overall charge should be that of the molecule.
- **For example**, one Mg ion has 2 positive charges and one Cl ion has 1 negative charge. Hence to have a total of zero charge we need 2Cl to combine with 1 Mg^{2+} ($\text{Mg}^{2+} + 2\text{Cl}$) to have a total charge of 0. Hence MgCl_2 , (not Mg_2Cl).

Cross and dot diagrams for ionic bonding

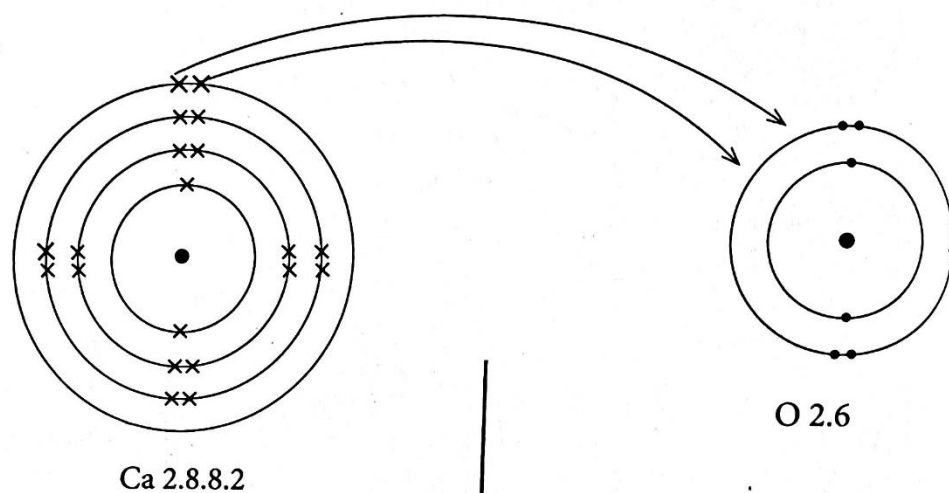
Examples: Show the ionic bonding of the following



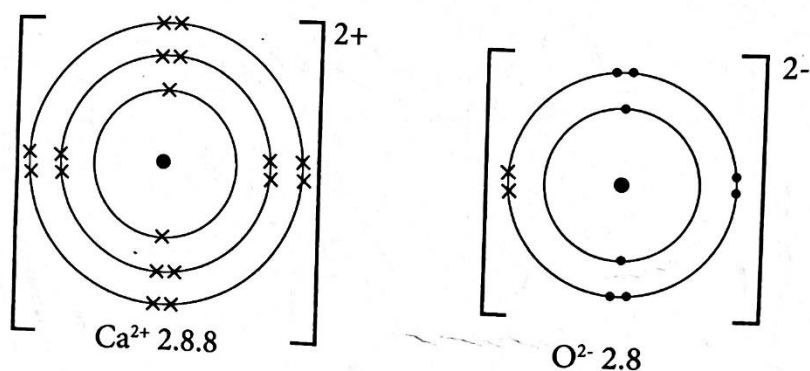
- The valence electrons from the magnesium atom are transferred to the outermost energy level of chlorine atoms as shown above.
- The formula for Magnesium Chloride is $\text{Mg}^{2+}\text{Cl}^-$ or MgCl_2

(b) Calcium oxide

Before electrons are transferred



After electrons are transferred



- The two valency electrons from one calcium atom are transferred to one oxygen atom.
- The formula of calcium oxide is $\text{Ca}^{2+}\text{O}^{2-}$ or CaO .

Properties of ionic bonding

- The following are some of the properties of ionic bonding
- They are soluble in water
- They have high melting and boiling point
- They are hard and brittle
- They are solid crystalline at room temperature
- They are insoluble in organic liquids like benzene or propanone
- They conduct electricity in molten or aqueous form

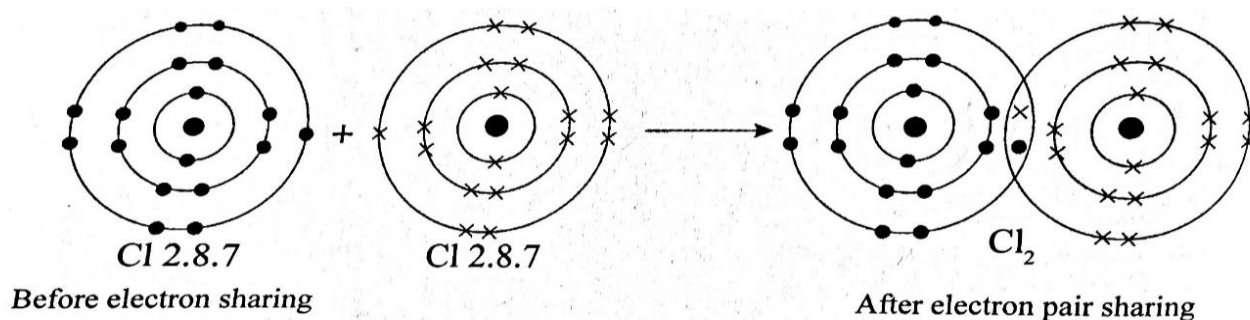
Covalent Bonding

- This is the type of bonding that involves sharing of valency electron/s between or among atoms.
- Is the bonding between non-metals.
- It involves the outermost shell electrons only.

Cross and dot diagram for covalent bonding

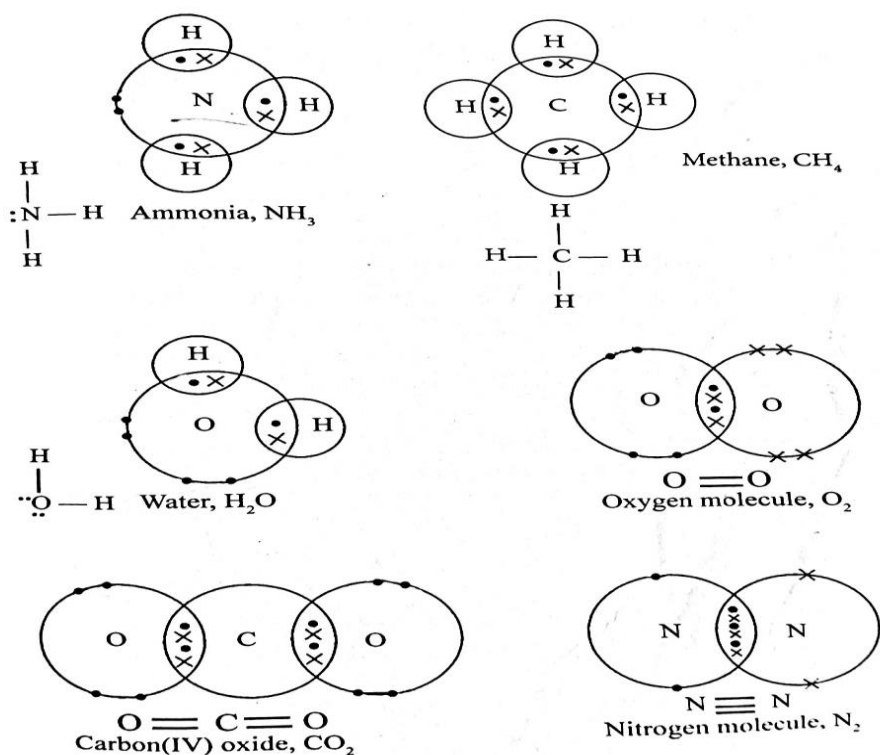
Examples:

a. Formation of Chloride molecule



- Both chlorines have shared an electron to make molecule stable

b. Formation of CH₄, O₂, H₂O, CO₂ and N₂

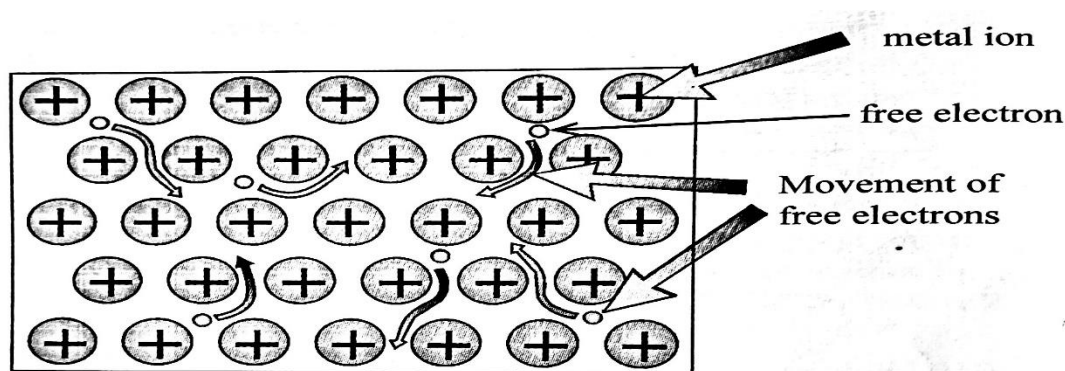


Properties of covalent bonding

- They have low intermolecular force.
- They have low melting and boiling points.
- They are volatile (can easily be vaporized) and have a particular smell for example Camphor.
- They are non-electrolytes.
- They are not very soluble in water.
- They are made up of molecules and not ions.

Metallic bonding

- It is the electrostatic attraction between fixed positive ions and free delocalized electrons that move and surround them.
- Below is the diagram of the metallic bonding how it takes place.



Physical properties of metals

- **Shiny.** They are used for jewelry and ornaments for example silver and gold.
- Good conductors of heat and electricity hence they are used for electrical wires for example copper.
- Hard wearing hence, they are used for bridge structures for example steel.
- Insoluble in water.
- Sonorous. They produce a ringing sound.
- Ductile. They are easily drawn into wires.
- Malleable. They are easily hammered into any shape.
- Metals have high melting and boiling points hence they are used for cooking
- They have high density.

Reasons for chemical bonding

- To attain stability.
- To attain low potential energy
- To experience more attractive forces.
- **Localised** refers to an object that is unstable, has high potential energy and experiences less attractive force.
- **Delocalised** refers to an object that is stable, has low potential energy and experiences more attractive forces.
- Hence the ions which are not combined are localized ions. those which are combined are called delocalized ions.

TOPIC 3: ACIDS AND BASES

Properties of acids and bases

Acids

- They have a sour taste for example: Lemons, Malambe, Oranges, sour milk
- They have a pH less than 7.
- They turn blue litmus paper red.
- They react with metals
- They react with bases
- They conduct electricity.
- They enhance rusting.

Electrical conductivity of acids

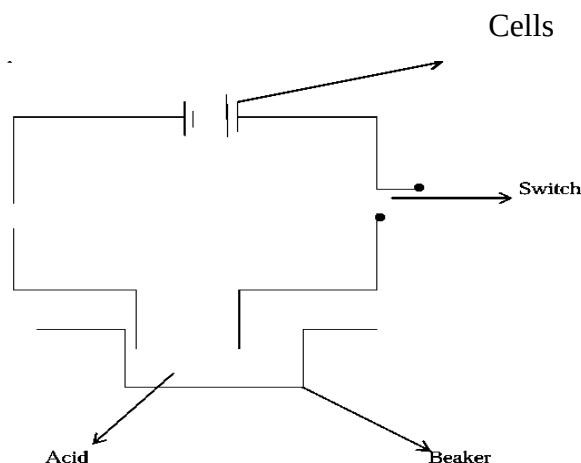
Experiment.

AIM: To show that acids conduct electricity

MATERIALS: Beaker, bulb, connecting wires, cells, switch, 2 different solutions (an acid and cooking oil or ethanol)

PROCEDURE

1. Set up the materials as shown below



2. Close the switch and note whether the bulb is giving light or not, and record your observation
3. Clean the beaker and replace it with another substance e.g. Ethanol and repeat step 2 above and note whether the bulb is giving light or not and record our observation

OBSERVATION

It will be noted that when there was acid in the beaker the bulb will give right while with ethanol the bulb will not give any right. The stronger the acid will result to the bulb giving more light

CONCLUSION

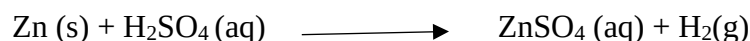
The experiment above shows that acid conducts electricity since it relies on hydrogen ions in water which drives electrons

Reactions of acids

The following are the reactions of acids.

Reaction with metals

Example Zinc (s) + Sulphuric acid (aq) $\longrightarrow$ zinc sulphate (aq) + Hydrogen (g)



Magnesium (s) + Hydrochloric acid(aq) $\longrightarrow$ magnesium chloride (+) Hydrogen(s)



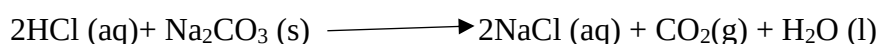
Reaction with bases to form salt and water

Example Hydrochloric acid (aq) + sodium hydroxide $\longrightarrow$ sodium chloride (aq) + water (l)



Reaction with metal carbonate and hydrogen to produce salt, carbon dioxide and water

Example Sodium hydrogen carbonate + hydrochloric acid $\longrightarrow$ sodium Chloride + carbon dioxide + water



Reaction with metal Oxide to form salt and water

Uses of acids

- Acids are used in car batteries for example sulphuric acid.
- They are used in the manufacture of paint, plastics, detergents, fertilizers.
- They are used in food preservatives.
- They aid in digestion for example hydrochloric acid.
- They are used in dyeing and electroplating for example methanoic acid.
- They are used as solvent for example ethanoic acid.
- They are used in plant growth.
- They are used for tanning of leather.

Bases

Properties of bases

- They have a bitter taste.
- They have a pH greater than 7.
- They turn red litmus paper blue.
- They react with acids.
- They conduct electricity.
- They enhance rusting.

Uses of bases

Sodium hydroxide

- (caustic soda) is used in the manufacture of soap.
- It is used in petroleum - refining, in making medicines and paper, pulp.
- It is used in making rayon in the textile industry.

Calcium hydroxide

- is also known as slaked **lime**.
- It is also used by farmers on the fields to neutralize acidic soil.
- It is used to manufacture of bleaching powder
- It is used as antidote for food poisoning.
- It is used in preparation of fungicides

Ammonium hydroxide

- Is used to remove ink spots from clothes and to remove grease from window - panes.
- Used in water purification.

Magnesium hydroxide

- Used as a laxative (stimulation of bowel movement)
- Used as an antacid in treatment of heart burn

Acid-Base Indicators

- Are all substances that can be used to show that the substance you have is an acid or base.

Types of indicators.

a. Commercial indicators

- These are indicators that are readily available and are easy stored in school laboratory
- Example are Litmus, phenolphthalein, methyl orange, screened methyl orange and bromothymol blue

Indicator	Colour in acid	Colour in base solution
Litmus paper	Red	Blue
Methyl-orange	Pink	yellow
Phenolphthalein	Colourless	Pink
Screened methyl-orange	Purple	orange

b. Locally made indicators

- These are indicators that can be prepared from local materials such as plant extract.

Preparation of acid-base indicators from selected leaves and flowers

- Preparation of acid - base indicator from Hibiscus flowers.

Experiment

Aim: to prepare acid - base indicator from Hibiscus flowers

Materials: hibiscus flowers, crucible, water, gas burner, vinegar, baking soda

Procedure:

1. Collect hibiscus flowers and cut into small pieces.
2. Place them in a hibiscus flowers crucible and add enough water to just cover the pieces.
3. Boil gently in a crucible for 10 minutes.
4. Cool the mixture.
5. Pour off the liquid, leaving the solid behind. Record the colour of the solution.
6. Add some of the solution to an equal volume of vinegar, shake and record the natural indicator colour in an acidic substance.
7. Add some of the solution to sodium bicarbonate (baking soda), shake and record the natural indicator colour in a basic substance.

8. If you obtained different colours for a solution with the acidic substance and with the basic substance, then you have prepared an acid - base indicator solution.

This solution will give one colour in contact with an acidic substance and another colour with a basic substance.

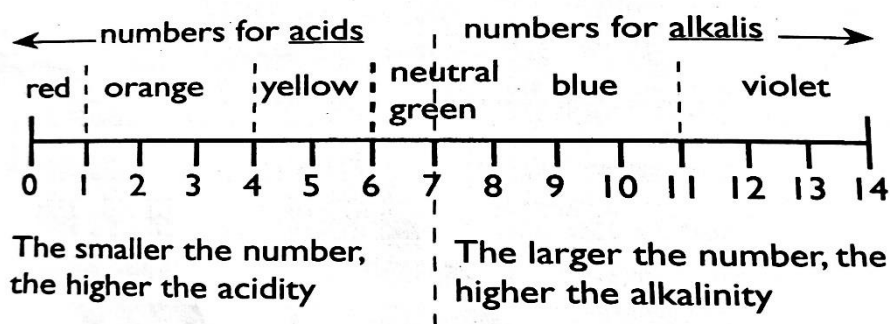
c. The pH scale and universal indicator

Universal indicator

- These are mixtures of dyes which gives a range of colours depending on strength of acid and base.
- It gives range of colours depending on the strength of the acid and alkali
- Universal indicator is a mixture of dyes. It gives a range of colours depending on the strength of the acid or alkali.
- Strongly acidic solutions turn universal indicator **bright red** for example, dilute sulphuric acid.
- Vinegar which contains ethanoic acid, is weakly acidic hence will turn universal indicator **orange - yellow**.
- Weak alkaline solutions turn it blue, while the strong alkaline solutions turn it violet.

The pH scale

- It is a scale of numbers that shows strength of an acid or a base.
- The strength of an acid or an alkali can also be shown using a scale of numbers called the pH scale.
- The scale runs from 1 to 14.
- When Universal indicator papers are used, a pH chart with the full range of colours and numbers on is used for comparison purposes.



- The strongest acid has a pH of 1

- The strongest alkali has a pH of 14
- If a substance is neutral, it has a pH 7

NOTE: Any solution less than pH 7 is acidic and any solution with a pH more than 7 is alkaline.

Neutralisation

- It is the act of making acid or base lose its acidity or basicity.
- It is done by reacting base and acid and the product is water and salt.

Application of neutralisation reaction

- Aluminium hydroxide, hydrogen carbonate, magnesium hydroxide is used to neutralize excess hydrochloric acid in the stomach.
- Calcium oxide (quicklime) or calcium hydroxide (slaked lime) is used to neutralize excess acids in acidic soils.
- Tooth paste for mouth cleaning has a base which neutralizes the acid which causes teeth decay.

TOPIC 4: HYDROCARBONS

Families of organic compound

- Hydrocarbons** are organic compounds that contain carbon and hydrogen only
- Oxycarbons** are organic compounds that contain carbon, hydrogen and oxygen

Homologous series

Is a group of compounds which have the similar properties and behave the same way.

Characteristics of homologous series

- The compound is represented by a certain general formula.
- They differ from one member to the next by a constant atom or group of atoms e.g. CH_2
- They can be prepared using similar methods
- They show gradual change in physical properties
- They exhibit similar chemical properties.

Examples of homologous series are:

- Alkanes alkanoic acids
- Alkenes alkynes
- Alkanols

Saturated and Unsaturated Hydrocarbons

- **Saturated** these are hydrocarbons that are formed when carbon bonds with three hydrogens in a chain.
- It has a maximum number of hydrogens bonded to it.
- **Unsaturated** these are hydrocarbons that are formed when carbon bonds with less than three hydrogens.

Alkanes

- Are hydrocarbons with the general formula $\text{C}_n\text{H}_{2n+2}$, where n denotes the number of carbon atoms in each molecule.
- The number, $n = 1, 2, 3, \dots$ etc.
- The general formula is used to find the molecular formula for the next alkane given number of carbons.
- **For example.**
 - Use the general formula of alkane to derive the molecular formula of alkanes with five carbons.

General formula: C_nH_{2n+2}

Given number of carbons is 5.

Substitute 5 for n into the general formula

$$C_{(5)}H_{(2 \times 5) + 2} = C_5H_{12}$$

The simplest alkane has only one carbon atom and it is Methane (CH_4).

Naming of alkanes

- The names of alkanes are derived by first writing a prefix and then adding the ending with “ane”

For example: One carbon atom is methane

Two carbons atom is ethane

Three carbons atom is propane

The four types of formulae of alkanes

a. Molecular formulae is the formula that shows the number of atoms involved in bonding.

b. Structural formulae is the formula that shows how atoms bond in the compounds in addition to giving the atoms involved in bonding.

c. Condensed formulae is the formula that categories atoms in groups depending on how they occur in structural formula.

d. The skeletal formulae is a zig-zag line representation of bonding in carbon atoms

Table showing the formula of the first ten straight chain alkanes

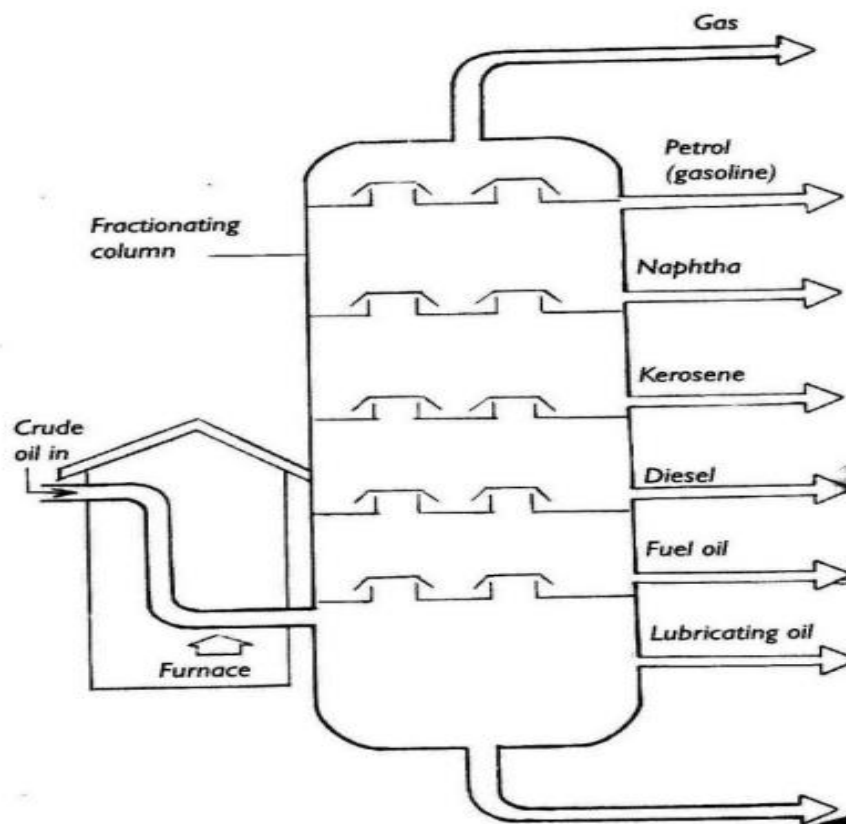
Number of carbon atoms	Name of alkane	Molecular formula	Structural formula	Condensed form of the structure
1	Methane	CH_4	$\begin{array}{c} H \\ \\ H - C - H \\ \\ H \end{array}$	CH_4
2	Ethane	C_2H_6	$\begin{array}{c} H \quad H \\ \quad \\ H - C - C - H \\ \quad \\ H \quad H \end{array}$	CH_3CH_3
3	Propane	C_3H_8	$\begin{array}{c} H \quad H \quad H \\ \quad \quad \\ H - C - C - C - H \\ \quad \quad \\ H \quad H \quad H \end{array}$	$CH_3CH_2CH_3$

4	Butane	C ₄ H ₁₀	$ \begin{array}{ccccccc} & H & H & & H & & H \\ & & & & & & \\ H & -C & -C & - & C & -C & -H \\ & & & & & & \\ & H & H & & H & & H \end{array} $	CH ₃ (CH ₂) ₂ CH ₃
5	Pentane	C ₅ H ₁₂	$ \begin{array}{ccccccccc} & H & H & & H & & H & & H \\ & & & & & & & & \\ H & -C & -C & - & C & -C & -C & -H \\ & & & & & & & & \\ & H & H & & H & & H & & H \end{array} $	CH ₃ (CH ₂) ₃ CH ₃
6	Hexane	C ₆ H ₁₄	$ \begin{array}{ccccccccc} & H & H & & H & & H & & H \\ & & & & & & & & \\ H & -C & -C & -C & -C & -C & -C & -H \\ & & & & & & & & \\ & H & H & & H & & H & & H \end{array} $	CH ₃ (CH ₂) ₄ CH ₃
7	Heptane	C ₇ H ₁₆	$ \begin{array}{ccccccccc} & H & H & & H & & H & & H \\ & & & & & & & & \\ H & -C & -C & -C & -C & -C & -C & -H \\ & & & & & & & & \\ & H & H & & H & & H & & H \end{array} $	CH ₃ (CH ₂) ₅ CH ₃
8	Octane	C ₈ H ₁₈	$ \begin{array}{ccccccccc} & H & H & & H & & H & & H \\ & & & & & & & & \\ H & -C & -C & -C & -C & -C & -C & -H \\ & & & & & & & & \\ & H & H & & H & & H & & H \end{array} $	CH ₃ (CH ₂) ₆ CH ₃
9	Nonane	C ₉ H ₂₀	$ \begin{array}{ccccccccc} & H & H & & H & & H & & H \\ & & & & & & & & \\ H & -C & -C & -C & -C & -C & -C & -H \\ & & & & & & & & \\ & H & H & & H & & H & & H \end{array} $	CH ₃ (CH ₂) ₇ CH ₃
10	Decane	C ₁₀ H ₂₂	$ \begin{array}{ccccccccc} & H & H & & H & & H & & H \\ & & & & & & & & \\ H & -C & -C & -C & -C & -C & -C & -H \\ & & & & & & & & \\ & H & H & & H & & H & & H \end{array} $	CH ₃ (CH ₂) ₈ CH ₃

Sources of alkanes

- Alkanes are mainly obtained through fractional distillation of crude oil.
- Crude oil is a complex mixture of hydrocarbons formed from dead plants and animals over millions of years.
- Alkanes can also be obtained from natural gas such as methane

Fractional distillation



Cracking of alkanes

- Cracking of alkanes is the process where by large alkanes are broken down into smaller, lighter and useful bites.
- The process involves heating of large alkanes using high presence of catalyst
- The process of heating large alkanes in the presence of catalyst is called catalytic cracking e. g Nonane cracking $\longrightarrow$ hexene + ethene + hydrogen

Physical properties of alkanes

- Melting and boiling points increase down the group.
- Are gases, liquids through solids at room temperature.
- Are insoluble in water.
- Are soluble inorganic solvents.
- Do not conduct electricity.
- Viscosity increases down the as the molecule becomes big, long chain alkane has high viscosity.

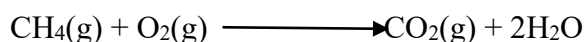
Chemical properties of alkenes

1. Alkanes undergoes both complete and incomplete combustion/reaction with oxygen

a) Complete combustion

- This is a reaction where there is sufficient supply of oxygen and produces carbon dioxide and water

Example Methane + oxygen $\longrightarrow$ carbon dioxide + water



b) Incomplete combustion

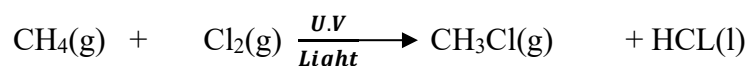
This is the reaction whereby there is insufficient supply of oxygen and the products are carbon monoxide and water

Example Methane + oxygen $\longrightarrow$ carbon monoxide + water

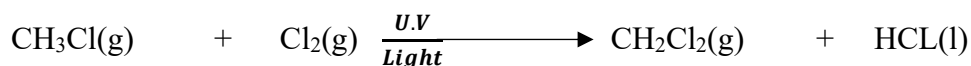
2. Alkanes undergoes substitution reaction with halogens

- The process is done by replacing hydrogen from an alkane with a halogen in the presence of **Ultraviolet light**. Eg methane reacting with chlorine in the following equations

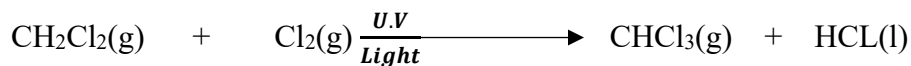
Methane + chlorine $\xrightarrow[\text{Light}]{\text{U.V}}$ **Chloromethane + Hydrogen Chloride**



Chloromethane + chlorine $\xrightarrow[\text{Light}]{\text{U.V}}$ **Dichloromethane + Hydrogen Chloride**



Dichloromethane + chlorine $\xrightarrow[\text{Light}]{\text{U.V}}$ **Trichloromethane + Hydrogen Chloride**



Trichloromethane + chlorine $\xrightarrow[\text{Light}]{\text{U.V}}$ **Tetrachloromethane + Hydrogen Chloride**



Uses of alkanes

- Methane is used as a source for cooking and fuel in vehicles.
- Propane and butane are used in school laboratories and industries for heating purposes.

- iii. Butane mixed with small proportions of propane are compressed in gas cylinders used in house hold lighting, water heating and cooking.
- iv. Petrol is used in automobile.
- v. Alkanes with higher numbers of carbons are used as solvent in the manufacture industrial chemicals.
- vi. Solid alkanes have others uses such as

Alkenes

- They are unsaturated hydrocarbons
- They have functional group of double bonds (C=C)
- The general formula for alkenes is C_nH_{2n}
- The smallest alkene is ethene (C_2H_4)

Naming of alkenes

- Alkenes are named by replacing “a” in an alkane with “e”

For example,

- Ethane gives ethene
- Propane gives propene
- Butane gives butene

Number of carbon atoms	Name of alkane	Molecular formula	Structural formula	Condensed form of the structure
2	Ethene	C_2H_4	<pre> H H C = C H H </pre>	$CH_2=CH_2$
3	Propene	C_3H_6	<pre> H H H C = C - C - H H H </pre>	$CH_2=CHCH_3$
4	Butene	C_4H_8	<pre> H H H H C = C - C - C - H H H H H </pre>	$CH_2=CHCH_2CH_3$

5	Pentene	C ₅ H ₁₀	$ \begin{array}{ccccccc} & H & H & & H & H & \\ & & & & & & \\ C & = & C & - & C & - & C & - & H \\ & & & & & & & \\ & H & & & H & H & H & \end{array} $	CH ₂ =CH(CH ₂) ₂ CH ₃
6	Hexene	C ₆ H ₁₂	$ \begin{array}{ccccccccc} & H & H & & H & H & H & H & \\ & & & & & & & & \\ C & = & C & - & C & - & C & - & C & - & H \\ & & & & & & & & \\ & H & & & H & H & H & H & \end{array} $	CH ₂ =CH(CH ₂) ₃ CH ₃
7	Heptene	C ₇ H ₁₄	$ \begin{array}{ccccccccccc} & H & H & & H & H & H & H & H & & \\ & & & & & & & & & & \\ C & = & C & - & C & - & C & - & C & - & C & - & H \\ & & & & & & & & & & \\ & H & & & H & H & H & H & H & & \end{array} $	CH ₂ =CH(CH ₂) ₄ CH ₃
8	Octene	C ₈ H ₁₆	$ \begin{array}{cccccccccccc} & H & H & & H & H & H & H & H & H & & \\ & & & & & & & & & & & \\ C & = & C & - & C & - & C & - & C & - & C & - & C & - & H \\ & & & & & & & & & & & \\ & H & & & H & H & H & H & H & H & & \end{array} $	CH ₂ =CH(CH ₂) ₅ CH ₃
9	Nonene	C ₉ H ₁₈	$ \begin{array}{ccccccccccccc} & H & H & & H & H & H & H & H & H & H & & \\ & & & & & & & & & & & & \\ C & = & C & - & C & - & C & - & C & - & C & - & C & - & C & - & H \\ & & & & & & & & & & & & \\ & H & & & H & H & H & H & H & H & H & & \end{array} $	CH ₂ =CH(CH ₂) ₆ CH ₃
10	Decene	C ₁₀ H ₂₀	$ \begin{array}{ccccccccccccc} & H & H & & H & H & H & H & H & H & H & H & & \\ & & & & & & & & & & & & & \\ C & = & C & - & C & - & C & - & C & - & C & - & C & - & C & - & H \\ & & & & & & & & & & & & & \\ & H & & & H & H & H & H & H & H & H & H & & \end{array} $	CH ₂ =CH(CH ₂) ₇ CH ₃

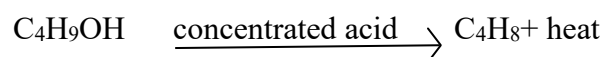
Sources of alkenes

a. *fractional distillation*

- The components are separated by fractional distillation. This is because the fractions have different boiling points.
- The crude oil is evaporated and its vapours allowed to condense at different temperatures in the fractionating column

b. *Dehydration of alcohol*

- Dehydration is the process of removing water from a compound.
- When water has been removed from alcohol, the result is alkene



- The final product is alkene called **butane**.

c. *Cracking of hydrocarbons*

- This involves the breaking of long chain alkane molecule into a shorter one and an alkene molecule.

- This is done under a high temperature.



Physical properties of alkenes

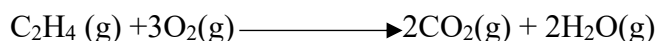
- Boiling and melting points increases as the molecule size increase.
- Density increases with an increase in molecule
- Are solid or liquid at room temperature.
- Are in soluble in water.
- Are soluble in organic solvents.

Chemical properties

a. *Can burn in air or oxygen. (Alkenes undergoes combustion)*

- This is the reaction of alkenes with oxygen. **For example,**

Ethene + oxygen $\longrightarrow$ carbon dioxide + water



b. *Reaction with hydrogen halides*

Ethene reacts with hydrogen chloride to form Chloroethane



c. *Under goes addition reactions*

- Addition of hydrogen also called **hydrogenation**.
- Ethane reacts with hydrogen in the presence of catalyst to form ethane.

d. *Hydration*

- When ethane is reacted with concentrated sulphuric acid it forms a compound called ethyl hydrogen sulphate.
- When added to water and warmed, an alcohol is formed called ethanol.

c. *Halogenation*

- Is the addition of hydrogen to an alkene
- This is done by substituting hydrogen with halogen atom.
- When the product is formed the prefix is borrowed from the name of the halogen where “ine” part is replaced by “o” e.g Bromo-alkane, Chloro-alkane, Iodo-alkane
- Halogen breaks the double bond of alkene.

Uses of alkenes

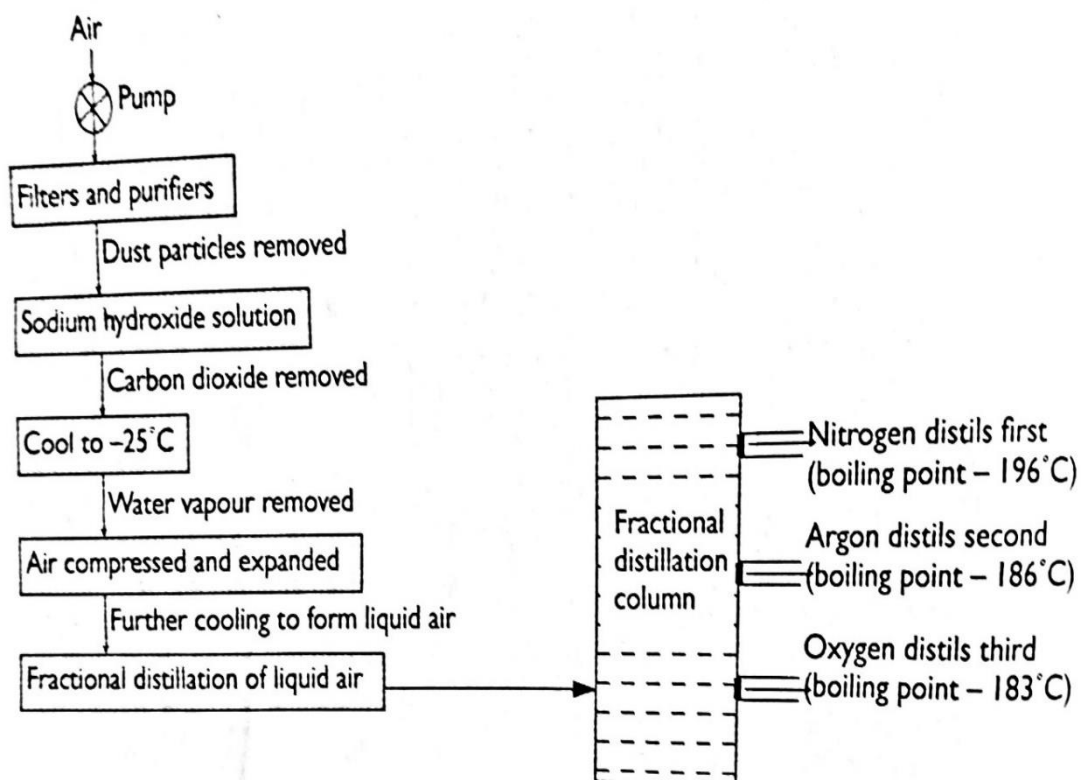
- Ethene and propene are used in the manufacture of plastics
- Ethene is used in artificial ripening of fruits.

TOPIC 5: AIR

What is air?

- Is the mixture of gases.
- Air contains the following
 - 78%nitrogen.
 - 21%oxygen.
 - 0.03%carbondioxide.
 - About 1noble gas
 - Variable smoke/dust particles
 - Variable polluting gases

Separating component of air using fractional distillation of liquid air



Uses of gases found in air

1. Nitrogen

- Dilute the oxygen in the air
- Weakens the process of burning and rusting in air
- Leguminous plants have bacteria which converts air nitrogen into useful compound rich in nitrogen.

2. Oxygen

- Is active fraction of air.
- Necessary for breathing.
- Used for burning.
- Used in rusting and decay.
- Without oxygen, there would be no life.

3. Carbon dioxide

- Air contains only 0.03%.
- Exhaled air is about 3%
- Some comes from burning wood, coal, petrol, oil and other compounds.
- Used in plants for photosynthesis.

4. Noble gases

- Examples include argon, neon, helium, krypton and xenon.
- Do not react with any other substance
- Argon is used in electric bulbs.
- Helium is used in weather balloons.
- Neon is used in coloured advertisement signs.

Quantitative determination of the fraction of oxygen in air

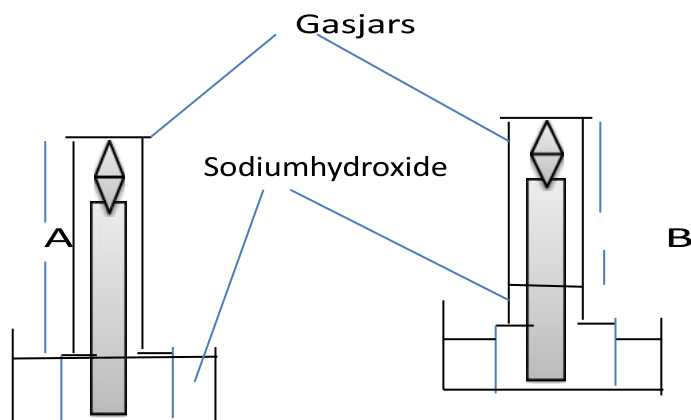
Aim: To determine the percentage of air used when a candle burns.

Apparatus: Trough, Bee hive shelf, Candle, Gas jar, 30cm ruler, Sodium hydroxide solution

Procedure:

- Invert an empty gas jar over the candle before lighting it.
- Measure the length of A in cm as shown in the figure below.
- Remove the gas jar.

- Light the candle and immediately cover it with the empty gas jar.
- When the candle flame goes off, measure height B in cm.



- When the candle is lit, it burns for some time then goes off.
- The water level rises inside the gas jar and goes down in the trough.
- Amount of air in the gas jar was A.
- Amount of air used during burning is A-B i.e. the rise in the level of water.

$$\text{Percentage of air} = \frac{\text{height A} - \text{height B}}{\text{height A}} \times 100\% = C\%$$

- Oxygen can also be prepared in the laboratory by reacting manganese oxide and hydrogen peroxide.
 - Hydrogen peroxide breaks into water and oxygen.
 - Manganese dioxide is used as a catalyst.
- Oxygen can also react with metals to produce rust under the presence of water.
- Oxygen can also cause burning of metals and other substances.

Physical properties of oxygen

- Is colourless gas.
- Has no smell
- Is slightly soluble in water.
- Is slightly denser than air
- Is neutral to litmus paper.

Chemical properties of oxygen

- Is very reactive.
- Supports burning.

Uses of oxygen

- In welding and cutting of metals.
- Used as aid to breathing where the natural supply is insufficient.
- Used to remove impurities in the process of making steel.
- Liquid oxygen is used to burn fuel in some space rockets.
- A mixture of charcoal, petrol and liquid oxygen is used as an explosive.

Atmospheric pollutants and their sources

- Atmospheric pollution is the introduction of substance to the environment which have harmful effects on the ability of the environment to support life.
- Can also be defined as making the environment dirty.
- Atmospheric pollution can be:
 - Poisonous gases
 - Solid particles

Poisonous gases and their effects on health and environment

a. Sulphur dioxide (SO₂)

- Are toxic and cause respiratory diseases.
- Also form acidic rain.

b. Nitrogen dioxide (NO₂)

- When nitrogen in the air combines with oxygen.
- Promotes respiratory diseases.
- Also forms acid rain

c. Carbon dioxide (CO₂)

- Prevents warm air from escaping into the atmosphere.
- Causes greenhouse effects. Green house effects lead to global warming.
- Causes the melting of polar ice caps which can raise the level of the oceans, hence lead to flooding.

d. Carbon monoxide (CO)

- From the burning fuels.

- It is a Poisonous gas.
- Can combine with hemoglobin in the red blood cells to form stable compound called carboxyhemoglobin.
- Causes suffocation and death
- Has no smell or colour which makes it difficult to be recognized

e. Unburnt hydrocarbons such as petrol fumes

- Can be carcinogenic.

f. Chlorine

- From decayed insecticides and perfume sprays.
- Destroys ozone gas.

g. Ozone (O₃)

- Generally, it's a light blue gas.
- Causes headaches among people.

h. Solid particles

- Large particles dumped by people in the environment.
- These may in water or on land.
- Examples include: Smoke from tobacco, Cement, Flour, Sugar, Plastic bags, Mining, Quarry.

Sources of pollutants

- Smoke from industries.
- Smoke from vehicles
- Charcoal burning
- Floods

Effort made to reduce atmospheric pollution

- Improvement of combustion of fuels.
- Introduction of better processing of fuels to make them free from Sulphur.
- Introducing more better filter systems in industries.
- Introducing of smokeless solid fuels or fuels like hydrogen.

TOPIC 6: SOIL

What is soil?

- Is a top material on the earth's crust that supports living.

Soil composition

Inorganic matter

- It is the 50% of the soil.
- Include rock particles from weathering, and minerals.
- These form soil particles such as: Sand, Silt, Clay
- Also form mineral elements such as: Iron, Aluminium, Copper, Silicates

Organic matter

- It is also called humus
- It is 5% of the soil.
- From animal wastes and human, decayed parts of plants and animals.

Water (Moisture)

- It is 25% of the soil.
- Is a solvent for minerals
- Too much water causes water logged soil.

Air

- It is 25% of the soil.
- Found in soil pores.
- Also called atmosphere or soil air.

Chemical composition of soil.

Cation exchangeable capacity

- Is the amount of exchangeable ions in 100g of soil. These include, Calcium, Potassium, Sodium
- Fertile soil has very high amount of exchangeable capacity (CEC).
- Soil with very high exchangeable sodium is toxic to most crops.

Soil salinity

- Is the amount of salts in the soil per unit mass
- Saline soils have excess salts especially sodium chloride which can be toxic.
- Most important chemical property of the soil.

Soil pH

- Determines the soil fertility in terms of mineral nutrients.
- Can be acidity or alkalinity.
- The degree of acidity is determined by the concentration of the hydrogen in the soil
- Strong acidic is 1 while weak acidic is 6.
- Strong alkaline is 14 and weak alkaline is
- 7 is neutral.

Soil pollution

- Can be polluted by

Oxidation of sulphur

- Cause acidic rains
- Also cause leaching of mineral nutrients leading to decrease in fertility of the soil.

Aerosols

- Are used to control pests and diseases in plants and animals.
- Usually contain heavy metals such as: Copper, Mercury. These are absorbed by plants.
- These plants are toxic when eaten by animals, can cause death.
They can also kill soil microorganism.

Petroleum products

- Oil spills kill soil microorganisms.

Agro-chemicals

- Include chemicals such as: Fertilizers, Insecticides, Pesticides, Fungicides

Solid wastes

- From households and industries.
- Some are biodegradable while other are non-biodegradable.
- Can cause injuries. Examples include, Glass bottles and Scrap metals.

Prevention of soil pollution

- Recycle all solid wastes
- Burn biodegradables wastes which can be burnt.
- Use organic farming to discourage excess use of chemical fertilizers
- Encourage using biological control of pests and diseases.
- Pipeline transport of petrol and petroleum products to minimize the risk of spilling.